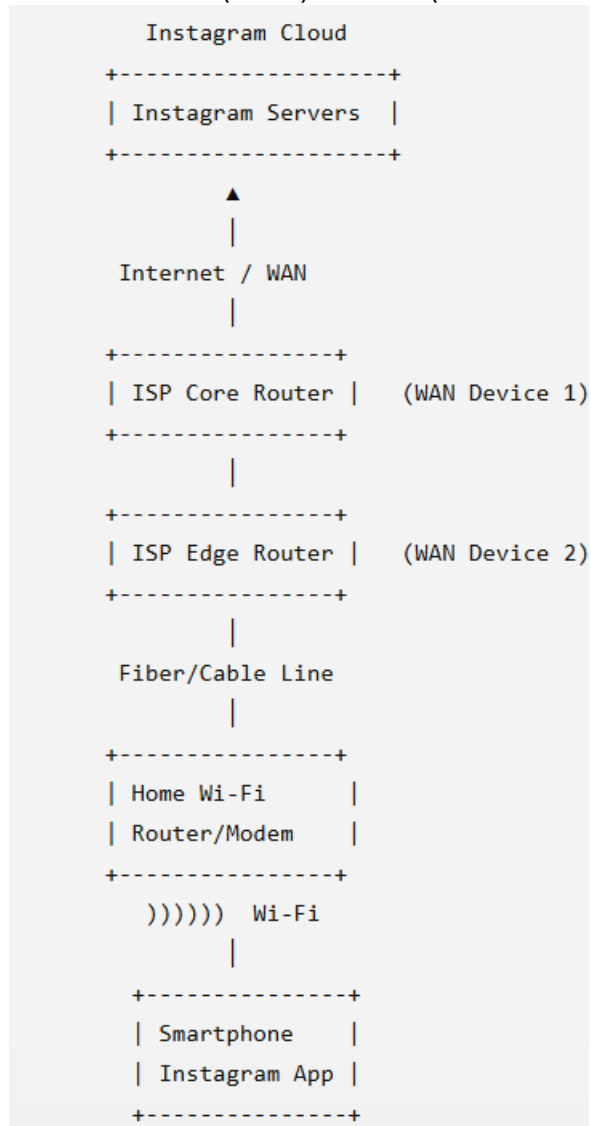


Wide Area Networks

Task 1.

Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.



Task 2.

Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

WAN Technology	Typical Speed	Cost	Real-Life Use
MPLS (Multiprotocol Label Switching)	High (100 Mbps–10 Gbps)	High	Connecting multiple Zomato offices and data centers securely
Leased Line	Very High (Dedicated 100 Mbps–10 Gbps)	Very High	IRCTC ticketing systems, banks, corporate headquarters
Broadband Internet	Medium to High (50 Mbps–1 Gbps)	Low	Home internet, small businesses, online learning

Task 3.

Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

Hint: Focus on how the VPN creates a private tunnel over the public WAN.

A Virtual Private Network (VPN) creates a secure, encrypted connection (called a private tunnel) between your device and a VPN server over the public Internet, which is a Wide Area Network (WAN).

For example, when you connect to the public Wi-Fi in a coffee shop and open Spotify or YouTube, anyone on the same Wi-Fi could potentially monitor unencrypted traffic. When a VPN is enabled:

1. Your device encrypts all internet traffic.
2. The encrypted data travels through the public WAN to the VPN server.
3. The VPN server decrypts the data and forwards it to Spotify or YouTube.
4. Responses travel back through the same encrypted tunnel.

This protects your data from hackers on public Wi-Fi and helps keep your browsing private.

Task 4.

Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Recommended WAN Technology: MPLS

Reasons:

1. **Reliable and Secure Connectivity:** MPLS provides secure and stable communication between warehouses in Ahmedabad, Mumbai, and Delhi, ensuring fast data transfer for inventory management and order processing.
2. **High Performance with Low Latency:** MPLS offers low latency and better Quality of Service (QoS), allowing real-time applications such as warehouse management systems, video conferencing, and logistics tracking to run smoothly.

Conclusion:

MPLS is the most suitable WAN technology for Flipkart because it provides secure, reliable, and high-performance connectivity between multiple warehouse locations.